Rule	Example
Rule 1 *Comments that are ambiguous are to be coded as not ED*	*"Hope you are having a good time up there… miss you forever… "* **(Not ED)**
Rule 2 *Ignore comment content without any emotional component*	*"Hey just had my first day of high school It was ok Still miss you "* **(First two sentences ignored.)**
Rule 3 *Memories of the deceased in and of themselves do not constitute ED*	*"I remember how great you were at ice hockey Whenever I play I know you are watching over me We sure are going to miss you "* **(First sentence disqualified.)**
Rule 4 *Common funeral styled and prosaic content does not qualify as ED*	*"I love you and thinking about you all the time We will never forget you "* **(All statements disqualified.)**
Rule 5 *Comments that include distressed language but that that is positively reframe the loss do not qualify as ED*	*"Thanks to your death I've gained a new appreciation for life I tried to live every day to the fullest like you always did Sometimes it still hurts to think of you and I know I will always miss you but I am still glad that I was fortunate enough to be in your life "* **(Not ED)**
Rule 6 *Comments that include references to ED in the past but do not include subsequent positive reframes are to be coded as ED*	*"This is the third anniversary of that terrible night I miss you more than ever we all do… for a while we could not even speak your name You are s0 loved and you will always be missed mom"* **(ED)**

Table 1 Summary of codebook rules for classifying post mortem comments as expressing emotional distress (ED)

non-ED as the default code: *"Comments that are ambiguous are to be coded as not ED."* For example, simple comments such as *"Miss you!"* were not coded as ED, while those unambiguously indicating distress were:

Happy 22nd Birthday Baby!! There are no words to describe how much I miss you, this day this weekend nothing is the same. The world for me has lost it's magic, it's beauty. I love you Matt-Matt there is not one single minute that passes that I am not thinking of you. Momma loves you baby!!!!!

Post-mortem comments often include non-emotive content, such as memories or personal updates. Rule #2 instructed coders to ignore content that does not have an emotional component, while rules #3 and #4 stated that memories of the deceased and common funerary styled content (*e.g., "We will never forget you.")* do not, in and of themselves, constitute ED. Rather, coders were instructed to focus on the interpretation and context of the content. Under these rules, the following comment would qualify as ED:

I miss you so much every day. I can barely live my life without you, I know I will miss you til the day I die

Whereas a similar comment would not:

We miss you a lot, but I know you are watching over us from heaven. Love you!

Finally, our coding was also sensitive to the ways in which the loss of a loved one can be reframed over time. Rule #5 specifies that if the comment includes content about how the deceased or the experience of their death enriched the author's life, it is unlikely that the comment is ED. For example, the following would not be coded as ED:

Thanks to your death I've gained a new appreciation for life. I tried to live every day to the fullest, like you always did. Sometimes it still hurts to think of you, and I know I will always miss you, but I am still glad that I was fortunate enough to be in your life

While this comment would qualify as ED:

Every day has been terrible since you've been gone. I feel like anything could just push me over the edge. I hold the memories of your smiles dear and I try to make my own like I know you would want me to, but I still can't think of you without the pain.

Finally, rule #6 specifies that comments discussing times in the past when the author was distressed, but that do not positively reframe the loss, can be coded as ED (subject to other rules). For example, the following message, while discussing the past, provides no subsequent reframe, and as such would be coded as ED:

When we were at your casket, Mom and I had to hold each other up. We cried so hard during most of the wake, and its still painful to think about. I love you and miss you very much.

Throughout, coders were instructed to weigh emotional content and give preference to the more potent and/or substantial aspects of the message.

Reliability Assessment

To assess inter-rater reliability, two of the authors independently coded 100 comments randomly selected from 10 deceased individuals' profiles. Inter-rater reliability on this coding was good with an agreement of 90% and kappa of .71. Next, 20 comments were coded by eight coders with 84.4% agreement. Upon the achievement of good inter-rater reliability, 2113 more comments were coded as ED or non- ED by one author. Our conservative definition resulted in low rates of occurrence with 12.5% of the comments including expressions of ED.

The Language of Emotional Distress

Having established a reliable way of distinguishing comments expressing emotional distress, our next objective was to determine whether linguistic patterns underlie expressions of emotional distress. In this part of our study, we aimed to find and articulate the linguistic style of emotional distress.

Linguistic style, how language is used by individuals or groups of people (Eckert 1996), provides information about the characteristics of the people using language as well as the conditions of their social surroundings. One way to quantitatively study linguistic style is to observe the use of linguistic features, such as parts of speech. This type of analysis focuses less on context and more on how language use varies between groups (see Pennebaker, Mehl, & Niederhoffer 2002 for review).

Previous research in both computer-mediated communication and traditional settings indicates that mental and physical well-being exhibit strong relationships with language use. Particularly, associations between certain parts of speech and emotional state have been observed in several previous studies. As reviewed by Pennebaker and his colleagues (*ibid*.), studies have reported that use of first person singular pronouns were elevated in the language use of individuals exhibiting signs of depression and suicide ideation. Researchers interpreted these heightened instances of pronoun use as high degree of self-preoccupation.

Patterns of word use bearing positive or negative sentiment, meanwhile, were also found to relate to health outcomes in the context of bereavement (Pennebaker *et al.* 1997). Specifically, higher rates of use of positive emotion words were linked to better physical and mental health.

Analysis of Language Use

To capture linguistic differences between ED and non-ED comments, we computed variables encapsulating the frequency of use of linguistic style features and sentiment expression.

In order to analyze the sentiment content as well and linguistic style of the comments in our dataset, we used "Language Inquiry Word Count" (LIWC)[2], a common language analysis package that provides dictionaries for parts of speech and punctuation, as well as psychological and social processes. LIWC has previously been used for analyzing social media content (*e.g.*, Getty *et al*. 2011; Kivran-Swaine & Naaman 2011). For each body of text analyzed, LIWC provides a score between 1 and 100 per category, specifying the proportion of words in the text contained in the dictionary for that category.

The *linguistic style* features we examined were pronouns, conjunctions, negations, adverbs, and tense use. Heightened use of first person pronouns and decreased use of second and third person pronouns have previously been found to relate to depression (Pennebaker, Mehl, & Niederhoffer 2002). Additionally, previous research suggests that use of first person singular pronouns elevate during emotional upheavals (*ibid*.). High use of adverbs and negations is an indicator of female style language, which is more socially aware (Mulac *et al*. 1998), while use of prepositions and conjunctions suggests a common frame of reference between the conversing parties. As such, uses of these parts of speech can be expected to elevate in more informal and intimate settings (Pennebaker, Mehl, & Niederhoffer 2002). Finally, tense use has been found to be an indicator of informality as well as need states in previous work (*ibid*.).

To assess *expression of sentiment*, we used scores from the following categories: positive emotion, sadness, anger, and social processes. The category, *positive emotion*, has been found to provide a basis for detecting expression of joy in social media content (Kivran-Swaine & Naaman, 2011), and includes words such as "happy", "love", "nice". *Sadness*, meanwhile, contains words such as "cry", "sad", and "grief", which all relate to the emotional state, sadness. Similarly, the category, *anger* contains words related to the emotional process of anger and aggression, such as "hate" and "kill". Finally, the *social processes* category contains social verbs such as "talk" as well as references to social relationships such as "family" and "friend."

For each comment in our dataset, we computed the LIWC scores for the following categories: word count, pronouns (first person singular, first person plural, second person, third person singular, third person plural), conjunctions, negations, adverbs, and tense use (present, future, and past tense), positive emotion, sadness, anger, and social processes.

Method

The unit of analysis in this work is a single comment. We first conducted multiple independent samples T tests to examine whether linguistic style and expression of sentiment differ between post-mortem comments that were coded ED or non-ED. The independent variables for the T-tests were the computed variables for each linguistic and sentiment category, and the grouping factor was the binary variable *ed*, (whether a comment was coded as ED). To check for the assumption of homogeneity of variances, we conducted Levene's tests. To account for our practice of conducting repeated tests looking at the mean differences between the same groups, we used the Bonferroni correction, adjusting significance levels to α/n when conducting n tests simultaneously.

Although independent-samples T tests give us information about the significance of mean differences, they do not account for higher order interactions between multiple language variables. Therefore, as the second step

[2] http://www.liwc net/liwcdescription.php

of our analysis, we constructed two binary logistic regression models. The first model, Model 1, uses measures of linguistic style variables while Model 2 uses measures of expressions of sentiment as independent variables. For both models, the binary variable *ed* is the response variable.

Results

The results from the independent samples T-Test indicate that linguistic style can be a strong marker for expression of ED. T-values, significance values (after the Bonferroni correction), and group-level descriptive statistics for the T-tests can be seen in Table 2. The odds ratios, significance levels, β values as well as standardized errors for the variables in binary logistic regression models can be seen in Table 3.

Greater use of first person singular pronouns, past tense verbs, adverbs, prepositions, conjunctions, and negations, as well as higher word count per comment were observed in ED comments; non-ED comments demonstrated higher levels of use of second person pronouns. Significant differences in sentiment content were also observed between ED and non-ED comments: non-ED comments showing higher use of social process words and words expressing positive emotion and sadness, and ED comments with higher use of anger words.

Model 1, explaining the tendency of a comment to be ED by looking at linguistic style variables, showed that higher word count, as well as higher use of first person singular pronouns, adverbs, conjunctions, and negations increase the likelihood of a comment to be classified as ED, whereas higher use of second person pronouns, third person singular and plural pronouns, and future tense decrease that likelihood. Model 2, which takes content features into account, showed that higher levels of use of positive emotion and sadness words decrease the likelihood of a comment to be ED, whereas higher use of anger words increases that likelihood.

Variable	t	df	p-value	d
Word Count	10 94	306 96	***	0 81
F P S Pronouns	9 05	569 37	***	0 45
S P Pronouns	5 08	645 75	***	0 24
Past Tense	4 73	461 57	***	0 25
Adverbs	8 61	493 53	***	0 46
Prepositions	6 6	533 25	***	0 34
Conjunctions	10 03	442 54	***	0 56
Negations	6 75	419 46	***	0 39
Social Proc	6 6	705 98	***	0 31
Pos Emotion	10 76	953 08	***	0 46
Anger	3 74	400 47	**	0 23
Sadness	3 6	1053 42	**	0 15
	ED Comments		**Non-ED Comments**	
	Mean	S D	Mean	S D
Word Count	122 4	102 3	53 57	62 03
F P S Pronouns	9 95	3 52	7 65	6 26
S P Pronouns	6 62	3 79	8 06	7 48
Past Tense	4 21	2 62	3 37	3 78
Adverbs	7 29	3 28	5 33	5 09
Prepositions	8 77	3 24	7 24	5 44
Conjunctions	6 59	3 01	4 55	4 13
Negations	1 88	1 68	1 13	2 15
Social Proc	14 17	5 45	16 94	11 54
Pos Emotion	4 19	2 76	6 7	7 2
Anger	0 46	0 87	0 24	1 04
Sadness	2 55	2 08	3 2	5 79
p< 1 * p< 05 ** p< 01 *** p< 001 **** p< 0001				

Table 2 ndependent Samples T Test Results and Descriptive Statistics

Variable	b	S.E.	p-value	O.R.
Model 1				
ntercept	4	0 32	****	
Word Count	0 01	0	****	1 01
F P S Pronouns	0 08	0 01	****	1 09
S P Pronouns	0 04	0 01	***	0 96
T P S Pronouns	0 14	0 07		0 87
T P P Pronouns	0 35	0 17	**	0 7
Future Tense	0 11	0 05	**	0 9
Adverbs	0 06	0 02	****	1 06
Conjunctions	0 08	0 02	****	1 08
Negations	0 15	0 03	****	1 16
Model 2				
ntercept	1 28	0 13	****	
Pos Emotion	0 09	0 02	****	0 91
Sadness	0 03	0 01	**	0 96
Anger	0 13	0 05	**	1 14
p< 1 * p< 05 ** p< 01 *** p< 001 **** p< 0001				

Table 3 Details for Binary Logistic Regression Models

Discussion

The results of the analysis of language use show that the sentiment and linguistic style of comments exhibiting ED differ significantly from other post-mortem comments.

ED comments on average are longer than non-ED comments; nearly double the length of non-ED comments. This difference may indicate that those experiencing ED spend more time constructing messages and in working through their feelings through writing. Word length for post-mortem writing is also influenced by the system into which it is posted. For example, Getty *et al*. (2011) reported an average of 39.9 words per message in a sample from Facebook, while Roberts & Vidal's (2000) analysis